

Supporting Information for “Attack-Method Choice Changes Robustness Conclusions for a Standard LSTM Rainfall-Runoff Model”

Y. Sun¹

¹College of Hydrology and Water Resources, Hohai University, Nanjing, China

Contents

1. Text S1: Model Training Configuration
2. Text S2: Basin Coverage and Experiment Counts
3. Text S3: Finite NSE Metric Handling
4. Table S1: Model Training Hyperparameters
5. Table S2: Basin Counts per Experiment Block
6. Table S3: Rerun Provenance Artifacts

Text S1. Model Training Configuration

The LSTM model was trained using the *neuralhydrology* framework with the configuration summarized in Table S1. The model uses five dynamic Daymet forcing variables and 14 static catchment attributes. Training was conducted for 30 epochs with a single-layer LSTM (128 hidden units), AdamW optimization, and NSE loss.

The adversarial study uses the local 531-basin temporal benchmark run stored under `results/05_adversarial_robustness/runs/reproduce_531_nse074_2025_1129_2145_ep30/`. The checkpoint is pinned to `model_epoch020.pt`; later saved checkpoints were not used for adversarial evaluation.

Table 1 — Model training hyperparameters.

Parameter	Value
Architecture	LSTM (single-layer)
Hidden units	128
Training epochs	30
Batch size	256
Sequence length	365 days
Optimizer	AdamW
Loss function	NSE
Dynamic inputs	Precipitation, $T_{\max}$, $T_{\min}$, shortwave radiation, vapor pressure
Static attributes	14 CAMELS attributes
Training period	01/10/1990–30/09/1995
Validation period	01/10/1995–30/09/2000
Test period	01/10/2000–30/09/2005
Random seed	159193

Text S2. Basin Coverage and Experiment Counts

The adversarial rerun uses the same 531 basin IDs as the temporal benchmark model. The same basin list is used for all adversarial experiment blocks; there is no separate basin-list merge from the previous result line.

After merging and deduplication, the full evaluation contains 17,523 individual experiment records across 531 unique basin IDs. Table S2 summarizes basin counts by experiment block.

Table 2 — Basin counts per experiment block.

Experiment block	Description	Catchments
Exp. 1	Attack comparison (5 methods $\times$ 4 ε values)	531
Exp. 2	Constraint ablation (3 constraints $\times$ 3 ε values)	531
Exp. 3	Targeted attacks (3 targets at $\varepsilon = 0.1$)	531
Exp. 4	Causal trigger (4 pre-event windows)	531
Exp. 5	C&W minimum perturbation	531
Merged total	Unique basin IDs across all experiment records	531
Merged total	Individual experiment records	17,523

Text S3. Finite NSE Metric Handling

All experiment cells cover the 531 benchmark basin IDs. However, NSE is undefined or unstable for evaluation windows with too few valid observations or near-constant observed discharge. The adversarial runner therefore selects windows satisfying more than 100 valid observations and normalized observed-flow standard deviation greater than 0.1. Seventeen basin IDs did not satisfy this variance requirement in any candidate window and therefore yield NaN values for $\text{NSE}_{\text{clean}}$, NSE_{adv} , and ΔNSE .

The affected basin IDs are: 04127997, 05120500, 06350000, 06404000, 06406000, 06409000, 06431500, 06447500, 06847900, 07142300, 07299670, 07301410, 08324000, 09306242, 09386900, 09447800, and 09484600. Raw streamflow and Daymet forcing files exist for these basins, and the test-period raw streamflow records are complete. The NaN values arise from the finite-metric guard, not from missing basin data. Main-text NSE headline statistics therefore use 514 finite basin metrics while provenance and experiment coverage remain 531 basin IDs.

References

Kratzert, F., Gauch, M., Nearing, G., & Klotz, D. (2022). NeuralHydrology—A Python library for deep learning research in hydrology. *Journal of Open Source Software*, 7(71), 4050.

Table 3 — Rerun provenance artifacts.

Artifact	Path or value
Victim run	<code>results/05_adversarial_robustness/runs/reproduce_531_nse074_2025_1129_2145_ep30/</code>
Pinned checkpoint	<code>model_epoch020.pt</code>
Adversarial config	<code>src/adversarial/configs/full_eval_531_epoch020.yaml</code>
Basin list	<code>src/adversarial/data/531_basins.txt</code>
Output root	<code>results/adversarial_eval/531_epoch020/</code>
Merged result file	<code>results/adversarial_eval/531_epoch020/merged_all.json</code>
Coverage audit	17,523 records; 531 unique basin IDs; no missing basin IDs
Finite NSE metrics	514 basins; 17 low-variance-window NaN basins